

Andres Barrios

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EDUCATION

The University of Texas at Austin

Bachelor of Science, Mechanical Engineering Honors

Certificate: Programming and Computation

May 2027

GPA: 4.0

TECHNICAL SKILLS

CAD/Analysis: NX, Teamcenter, SolidWorks, Inventor, Ansys Mechanical, DFMA, Tolerance Stack-Up Analysis

Manufacturing: 3D Printing, CNC Machining, Injection-Molding, GD&T

Programming: Python, MATLAB, SQL, MongoDB, Neo4j

PROFESSIONAL EXPERIENCE

Apple – Product Design Engineer Intern | Cupertino, CA

January 2026 - July 2026

- Executed iterative experiments evaluating swaging and specialized crimping strategies incorporating first principles to redistribute crimp material, isolating viable paths to round and earless crimp cross-sections
- Analyzed vendor crimping settings against in-house capabilities to identify critical crimp die features, compression behavior, and process conditions for replication in internal lab prototyping
- Designed and prototyped custom crimp dies and an adjustable cable-holding fixture using NX, enabling repeatable cable-to-die positioning and isolated study of die effects on ear formation
- Synthesized findings into design rules of thumb identifying crimp wall-thickness ratio and zero-gap die closure as key drivers of earless crimping, presenting recommendations to guide adoption across design team

Samsung – Mechanical Design Engineer Intern | Austin, TX

May 2025 - August 2025

- Designed mechanical assembly in Autodesk Inventor implementing electric actuators and custom-built pipe collar to safely lift 200 lb piping within pump input subsystem, eliminating operator handling risks
- Created detailed GD&T drawings for machined pipe collar and adjustable claw clamps, using manufacturer DFM feedback to optimize parts for machining while upholding existing design constraints
- Analyzed failure modes of lifting system, confirming via hand calculations that clamping-induced hoop stress remained below pipe yield strength and frictional torque resisted actuator-induced rotation

Amazon Robotics – Hardware Development Engineer Intern | Boston, MA

January 2025 - May 2025

- Operated within team of mechanical engineers in design development cycle of new autonomous drive unit robot releasing in 1000+ distribution centers, including prototyping and validation of hardware
- Designed bolted wheel hub and taper-lock attachment in SolidWorks for drive unit, aimed to facilitate wheel removal from locomotors and reduce downtime during testing and inspection by over 50%
- Optimized wheel system assembly by performing tolerance stack-up analyses and updating GD&T drawings for both machined parts, eliminating worst-case interference between critical wheel and locomotor surfaces
- Prototyped six modular drive-units, enabling in-facility testing of locomotor and sensor field-of-vision mapping

BP – Mechanical Reliability Engineer Intern | Whiting, IN

May 2024 - August 2024

- Developed injection quill design to minimize fluid impingement at pipe tee junction, analyzing flow conditions to optimize material selection and evaluate vibration effects on quill stability in piping system
- Designed new drainpipe for condensate removal, improving steam heat transfer efficiency to piping systems by creating new drainage line to prevent condensate accumulation in critical fluid lines

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

ASCEND Texas – Mechanical Design Engineer

September 2026 - Present

- Designing mechanical components for autonomous paraglider ATV used for emergency and disaster response

Management Leadership for Tomorrow (MLT) – Career Preparation Fellow

January 2025 - Present

- Accepted into selective 18-month professional development program for high-achieving talent
- Complete case studies and assignments to grow leadership and technical skills relevant to the technology sector
- Attend conferences hosted by industry leaders such as Apple, Google, and LinkedIn